(Please write your Enrollment Number) Enrollment No.

END TERM EXAMINATION
(CBCS)(SUBJECTIVE TYPE)
B. Tech (IT/MAE/DMAM), 2nd Semester
(AUGUST, 2022)

Subject Code: BAS-106	Subject: Environmental Sciences
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			**(2.5*8=20)**
	(a)	Distinguish between undernutrition and malnutrition. What are the common diseases due to malnutrion?	
	(b)	Discuss advantages and disadvantages of physio-chemical treatment of waste water as opposed to biological treatment.	
	(c)	Explain the degradation of photo-degradable polymers on the basis of chemical agents and physical factors.	
	(d)	Out of aromatics, branched chain and straight chain hydrocarbons in a fuel for an internal combustion engine, which one would have better anti-knock characteristics and for which type of engine?	
	(e)	Explain the concept of carbon credits with its merits and demerits?	
	(f)	Which form of Chromium is essential for human body and why? What are the health hazards and biochemical effects associated with different forms of Chromium?	
	(g)	Write two chemical reactions where atom economy of more than 80% is recorded. Calculate the atom economy of producing CO2 by burning propane.	
	(h)	What is Geothermal energy? What are its merits and demerits?	
		UNIT-I	
Q2	(i)	Discuss the impacts due to deforestation activities and factors that have contributed to forest conservation in India.	**(5*2)**
	(j)	What is meant by Environmental Impact Assessment? Why is it necessary to have EIA before the development of any project?	
Q3	Write short notes on: (i) Impact of mining activities on environment (ii) Land subsidence (iii) Water harvesting (iv) desertification		**(2.5*4)**
		UNIT-II	
Q4	**(a)**	Explain the Physical and Chemical processes used for the treatment of hazardous waste.	**(5*2)**
	(b)	Discuss the 'selection of renewable feedstocks as starting material' and 'Zero Waste Technology' with suitable examples	
Q5	(a)	What is Photochemical Smog? Discuss the chemistry involved in its formation.	**(5*2)**
	(b)	Discuss (i) sources of soil pollution (ii) disposal methods for nuclear waste	

		UNIT-III	
Q6	(a)	A sample of coal containing 92% C, 5% H, 3% ash. When this coal was tested for the calorific value in the bomb calorimeter, the following data were obtained:- Weight of coal burnt = 0.95 gm, Weight of water taken = 700 gm, Water equivalent of bomb and calorimeter = 2,000 gm, Rise in temperature = 2.48 °C, Cooling correction = 0.02 °C, Fuse wire correction = 10.0 cal, Acid correction = 60.0 cal. Calculate the net and gross calorific value of the coal in cal/gm.	**(5*2)**
	(b)	What is Power Alcohol? Discuss its method of preparation.	
Q7	**(a)**	List various renewable sources of energy and explain merits and limitations of ocean thermal energy.	**(5*2)**
	(b)	A sample of coal was found to contain the following: C = 81 %, H = 4 %, O = 2 %, N = 10 %, remaining being ash. (i) Calculate HCV and LCV of 1 kg of coal sample, (ii) Estimate the quantity of minimum air required for complete combustion of 1 kg of sample.	
		UNIT-IV	
Q8	(a)	Enumerate various forms of Mercury and describe their sources and biochemical effects.	**(5*2)**
	(b)	What are Bioplastics? Explain methods of manufacture of Bioplastics.	
Q9	(a)	Write short notes on the following giving suitable examples (i) Acute and chronic toxicity (ii) Impact of toxic metals on the enzymes	**(5*2)**
	(b)	Calculate the number average and weight average molecular masses of polypropylene polymer with the following composition: $[-CH_2-CH(CH_3)-]_{100}$ is 40 % and $[-CH_2-CH(CH_3)-]_{250}$ is 60 % {Given that atomic mass of C = 12, H = 1}	